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EMPLOYEE ATTRITION PREDICTION USING MACHINE LEARNING

UNDERSTANDING WORKFORCE PATTERNS THROUGH DATA

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Project: Employee Attrition Prediction using Machine Learning

Tools Used: Python | Pandas | Seaborn | Matplotlib | Scikit-Learn

Dataset Size: 1470 Employees | 31 Features

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EXECUTIVE SUMMARY

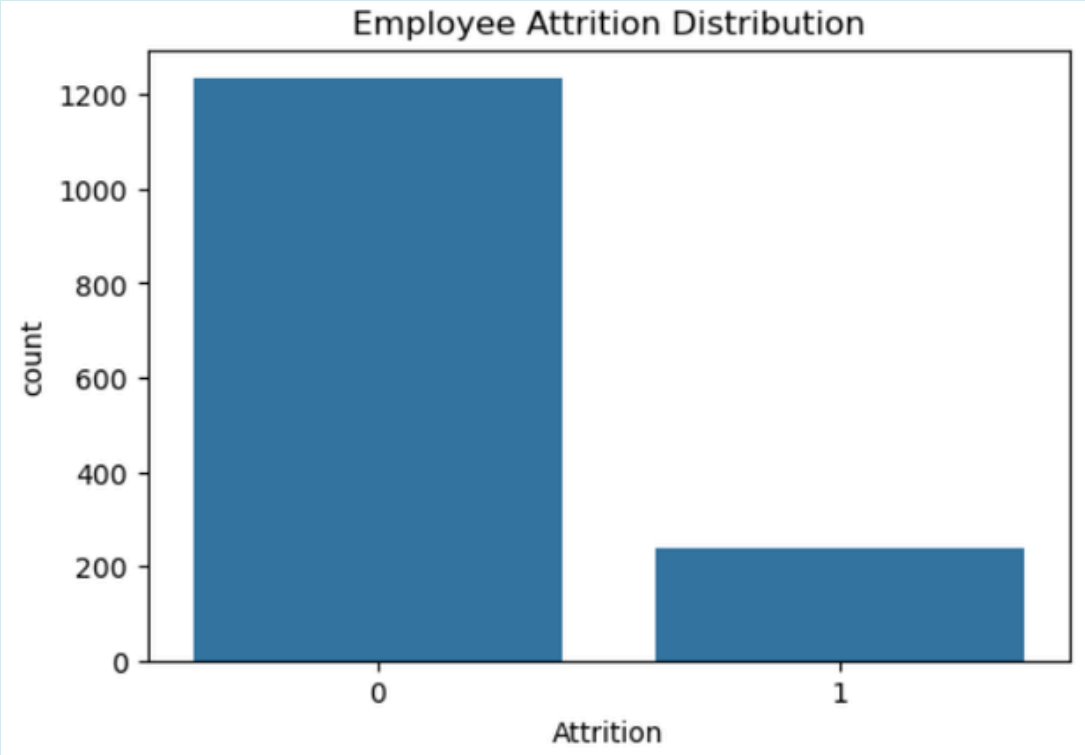
This exploratory data analysis examines employee data to identify key factors influencing attrition. The analysis reveals that overtime workload, job roles, career level, and work-life balance significantly influence employee turnover.

Key Findings:

- Employees working overtime show higher attrition
- Entry-level employees leave more frequently
- Sales roles show higher turnover
- Work-life balance strongly impacts retention
- Career growth influences employee decisions

Business Impact:

The insights from this analysis can help organizations design better employee retention strategies and reduce workforce turnover costs.



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DATASET OVERVIEW

Metric	Value
Total Employees	1470
Total Features	31
Numerical Features	26
Categorical Features	9
Target Variable	Attrition

Data Preparation Summary:

- No missing values found
- No duplicate records detected
- Removed constant features:
  - EmployeeCount
  - StandardHours
  - Over18
- Removed identifier column:
  - EmployeeNumber

Data Status:

Dataset is clean and ready for analysis.

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EMPLOYEE ATTRITION OVERVIEW

The first step was understanding how many employees left versus stayed.

Key Observations:

- Majority employees remain in company
- Minority employees leave organization
- Attrition rate approximately 16%

Key Insight:

Employee attrition is relatively low but still significant enough to impact organizational performance.

Business Interpretation:

Even moderate attrition can lead to:

- Hiring costs
- Productivity loss
- Training expenses

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WORK CONDITION IMPACT ON ATTRITION

Work-related variables show strong influence on employee turnover.

Overtime Analysis:

Employees working overtime show significantly higher attrition compared to those with balanced workloads.

Business Travel Analysis:

Frequent business travel shows higher attrition compared to rare travel.

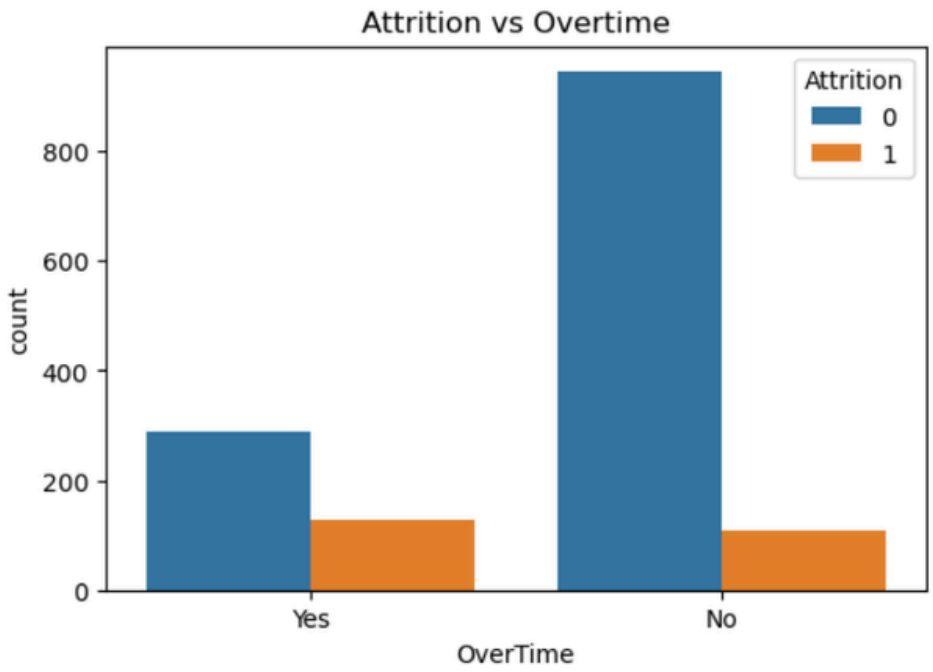
Key Insight:

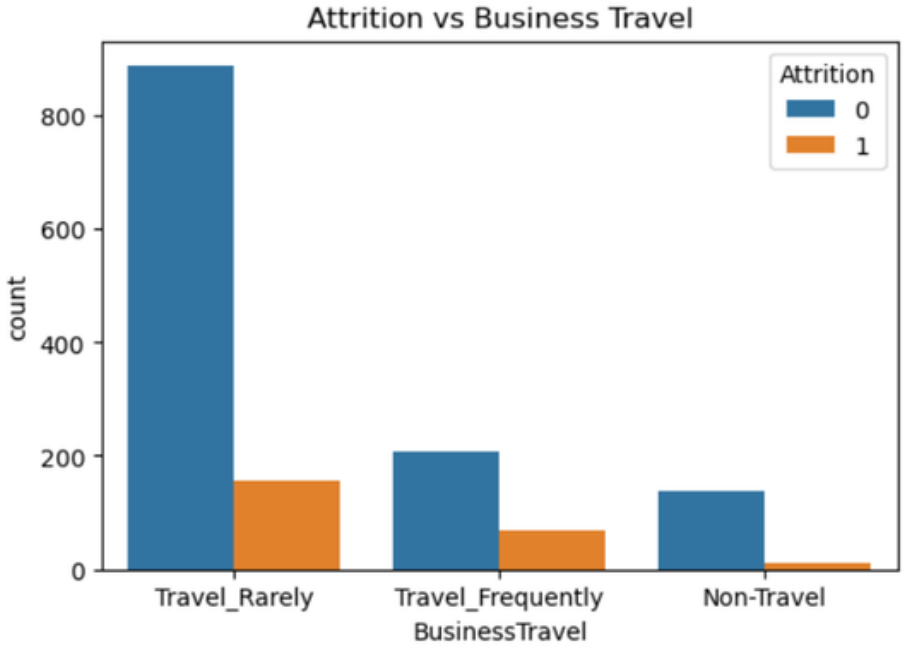
Work stress and workload are major contributors to employee turnover.

Business Recommendation:

Organizations should:

- Monitor employee workload
- Improve work-life balance
- Reduce excessive overtime





JOB ROLE BASED ATTRITION PATTERNS

Attrition varies significantly across job roles.

Observations:

- Sales Executives show higher attrition
- Laboratory Technicians show higher turnover
- Manager roles show lowest attrition

Key Insight:

Entry-level and operational roles experience higher employee turnover.

Business Interpretation:

Possible causes:

- Limited career growth
- Work pressure
- Compensation expectations

Recommendation:

Organizations should strengthen retention programs for high-risk roles.

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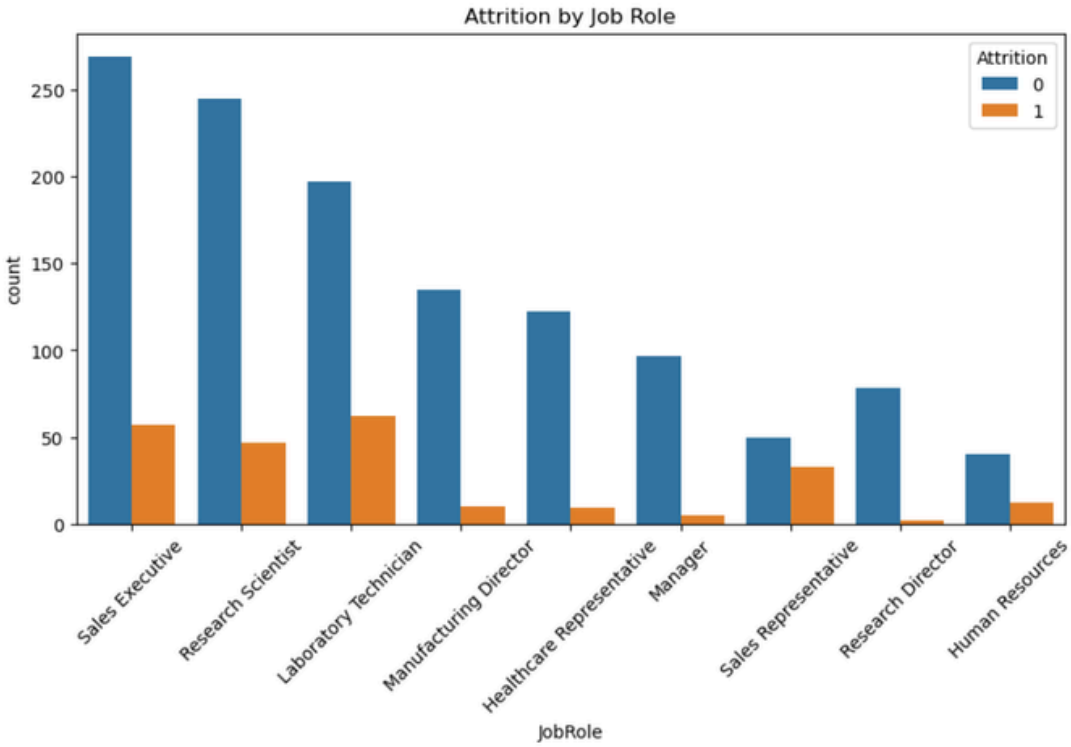

DEMOGRAPHIC INFLUENCE ON ATTRITION

Employee personal factors also influence attrition patterns.

Marital Status:
Single employees show higher attrition compared to married employees.
Age Factor:
Younger employees tend to switch jobs more frequently.
Gender:
Attrition difference between genders is minimal.

Key Insight:
Career stage and personal stability influence employee retention.

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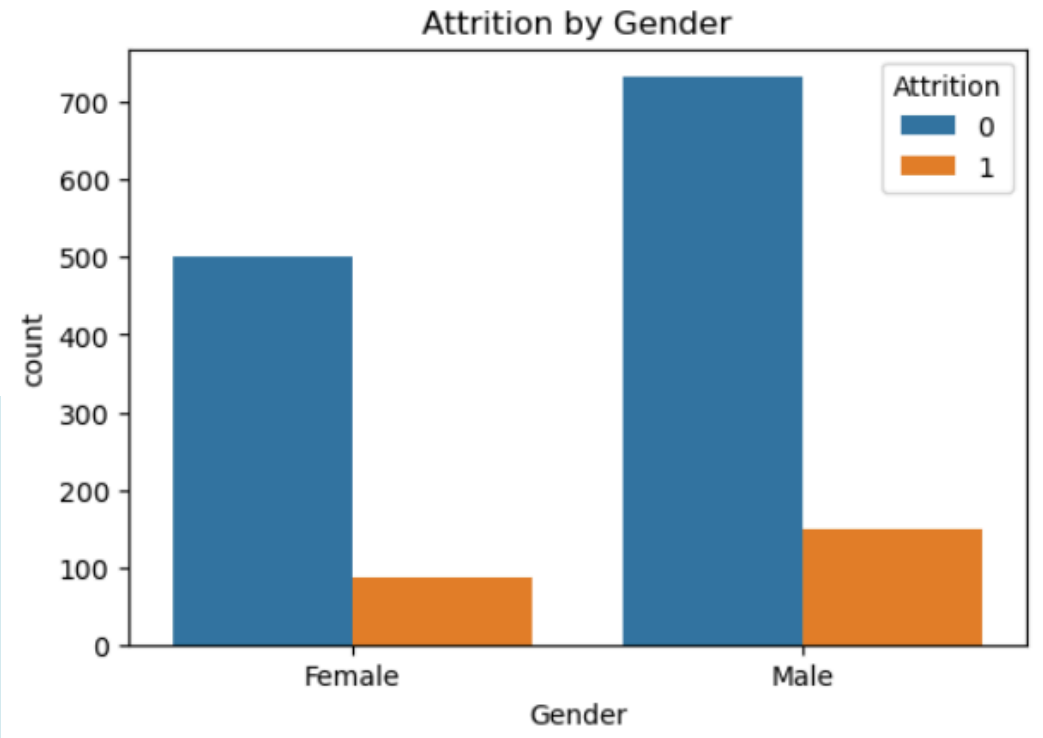
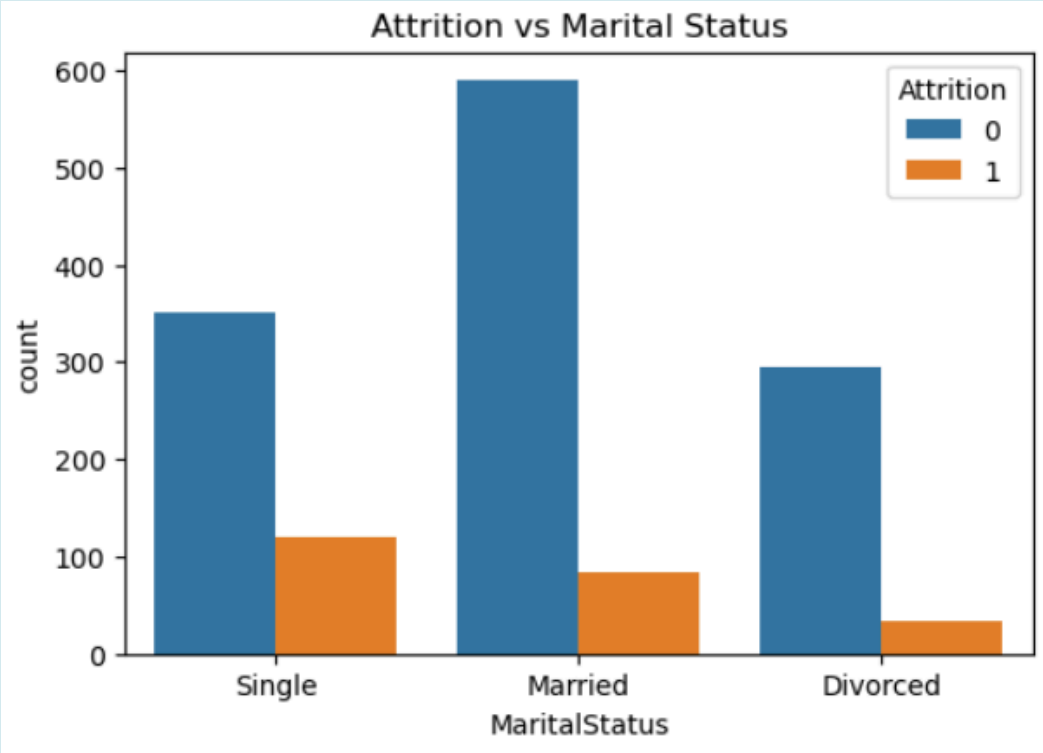
Recommendation:

Organizations should:

- Provide growth opportunities for younger employees
- Improve engagement programs





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KEY INSIGHTS FROM EXPLORATORY DATA ANALYSIS

Based on the analysis, the following actions are recommended:

Workforce Management
Reduce excessive overtime workloads.

Career Development
Provide clear career growth paths.

Employee Engagement
Improve employee satisfaction programs.

Retention Programs
Focus on high attrition departments.

Work-Life Balance
Encourage flexible work policies.

STRATEGIC RECOMMENDATIONS

INSIGHT 1 — OVERTIME RISK
EMPLOYEES WORKING OVERTIME SHOW SIGNIFICANTLY HIGHER ATTRITION RISK.

INSIGHT 2 — CAREER LEVEL RISK
LOWER JOB LEVELS SHOW HIGHER TURNOVER RATES.

INSIGHT 3 — ROLE RISK
SALES AND TECHNICAL ROLES SHOW HIGHER ATTRITION.

INSIGHT 4 — WORK-LIFE BALANCE
POOR WORK-LIFE BALANCE INCREASES ATTRITION.

INSIGHT 5 — EXPERIENCE FACTOR
EARLY CAREER EMPLOYEES CHANGE JOBS MORE FREQUENTLY.

APPLICATION IN MACHINE LEARNING MODEL

The insights obtained from exploratory data analysis will be used to select important features for machine learning models.

Important Features Identified:

- OverTime
- JobRole
- JobLevel
- MonthlyIncome
- YearsAtCompany
- WorkLifeBalance

Next Steps:

- Feature encoding
- Model training
- ROC evaluation
- Model comparison

Project Objective:

Build a predictive model to identify employees likely to leave.

This demonstrates practical application of HR analytics using machine learning.

CONCLUSION

THE EXPLORATORY DATA ANALYSIS REVEALED IMPORTANT WORKFORCE PATTERNS AFFECTING EMPLOYEE ATTRITION. WORKLOAD, JOB ROLE, CAREER LEVEL, AND WORK-LIFE BALANCE EMERGED AS MAJOR FACTORS INFLUENCING EMPLOYEE DECISIONS.

THESE INSIGHTS PROVIDE VALUABLE DIRECTION FOR BUILDING PREDICTIVE MODELS AND DESIGNING WORKFORCE RETENTION STRATEGIES.

HR ANALYTICS COMBINED WITH MACHINE LEARNING ENABLES ORGANIZATIONS TO MOVE FROM REACTIVE HR MANAGEMENT TO PROACTIVE WORKFORCE PLANNING.